

VOLUME OF PRISMS

A line has only one dimension—length. A flat shape has two dimensions—length and width. Most objects in life have three dimensions—length, width, and height. Depending on the situation, one of these dimensions may be called *depth*.

A **prism** is a three-dimensional object with parallel, identical bases that have straight edges. The sides, or faces, that connect the bases are always rectangles.

The drawing shows a **rectangular prism** and a **triangular prism**. Notice that the base determines the type of prism.

∴ ALL SIDES ARE "FACES" BUT ONLY PARALLEL SIDES ARE "BASES."

A corner of the prism is a **vertex**. The **edges** are the straight segments that define the prism's shape. The flat side of a prism is also called its **face**.

In the drawing, the prism has **8 vertices**, **12 edges**, and **6 faces**. Take a moment to find them all.

Volume is the measure of the space inside a three-dimensional object. Volume is measured using **cubic units**. To find volume, you need to figure out how many cubes will fit inside the figure. Use the formula below to find the volume of any prism.

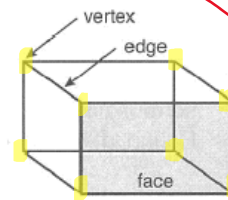
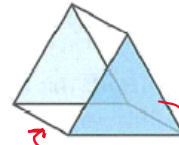
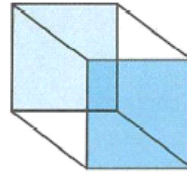
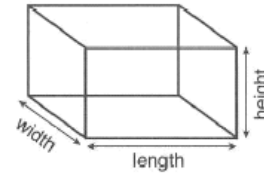
Volume (V) of a Prism

$V = Bh$, where **B** is the area of one base and **h** is the height

any one of these dimensions can be called

DEPTH—

Just as a rectangle can L & W OR L & H OR W & H, etc.



the 2 Δs are the Bases Because parallel & identical the 2 Faces are rectangles

Examples #1

Saturday, August 10, 2024 7:22 PM

Volume (V) of a Prism

$V = Bh$, where B is the area of one base and h is the height

FINDING THE VOLUME OF A PRISM

Example Find the volume of the prism.

Step 1

Identify the bases.

Remember, bases are identical and parallel. The bases are right triangles.

Step 2

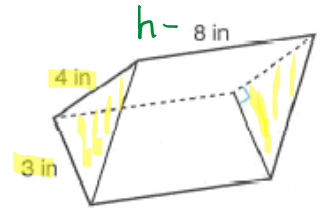
Find the area of one base.

$$\begin{aligned} A &= \frac{1}{2}bh \\ &= \frac{1}{2} \cdot 4 \cdot 3 \\ &= 6 \text{ sq in} \end{aligned}$$

Step 3

Multiply by the height, 8 inches.

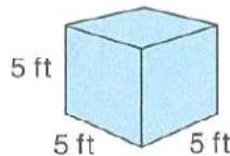
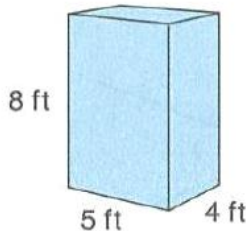
$$6 \cdot 8 = 48 \text{ cubic inches or } 48 \text{ in}^3$$



The Δ sides are the Bases because they are the Faces equal & parallel.

7. How much greater is the volume of the first box than the volume of the second box?

$$V = Lwh$$



Since Rectangular Prisms, any pair of equal & parallel Faces can be the bases.

$$V = (8)(5)(4) \quad V = 5^3$$

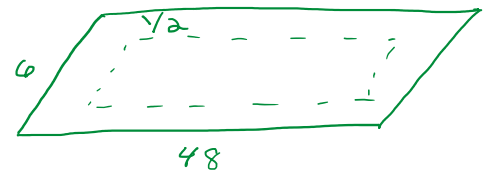
$$V = 160 \text{ ft}^3 \quad V = 125 \text{ ft}^3$$

Difference in Volumes $160 - 125 = 35 \text{ ft}^3$
 what % of 125 is 35? $35/125 = 28\%$

8. A plan for a concrete wall calls for it to be $\frac{1}{2}$ foot thick, 6 feet high, and 48 feet long.

a. What is the volume of the wall in cubic feet?

b. Concrete is sold in cubic yards. One cubic yard = 27 cubic feet. How many cubic yards of concrete are needed to build the wall?



$$a) V = (6)(48)(\frac{1}{2})$$

$$V = 144 \text{ ft}^3$$

$$b) 144/27 = 5.\bar{3}$$

$\therefore 6 \text{ cubic yards}$

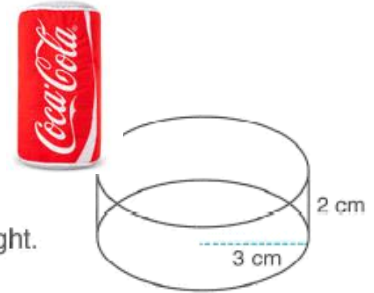
Volume of Cylinders

Saturday, August 10, 2024 7:23 PM

A **cylinder** is a **three-dimensional** shape with **circular bases**. The **two bases** are **identical and parallel**.

Like a coke can.

To find the volume of a cylinder, find the area of one base and multiply by the height.



The GED math formula sheet includes the formula for volume of a cylinder, which multiplies the area of one circular base (πr^2) by the height h . *you don't need this!!*

Volume (V) of a Cylinder $V = \pi r^2 h$, where r = radius and h = height

*Base is a circle with Area = πr^2 .
Then cut open the cylinder, lie it flat, and you have a rectangle with height h .
Don't memorize unnecessary formulas!!*

Example Find the volume of a cylinder with a **radius of 3** centimeters and a **height of 2** centimeters.

$$\begin{aligned} V &= \pi r^2 h \\ &= 3.14 \cdot 3^2 \cdot 2 \\ &\approx 56.52 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{area of its Base } A &= \pi r^2 \\ &= 3.14 (9) \\ A &= 28.26 \end{aligned}$$

The cylinder has a volume of about **56.52 cubic centimeters** or **56.52 cm³**.

$$\begin{aligned} \text{Volume} &= (\text{area of its Base}) (\text{height}) \\ &= 28.26 (2) \end{aligned}$$

$$V = 56.52 \text{ cm}^3$$

a

Examples #2

Saturday, August 10, 2024 7:26 PM

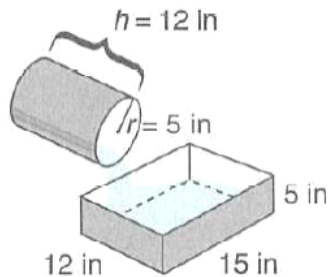
9. A cylindrical fuel tank has a height of 20 feet and a diameter of 16 feet. What is the volume of the fuel tank in cubic feet?

$$\begin{aligned}h &= 20 & A_{\text{base}} &= \pi r^2 \\r &= 8 & &= (3.14)(8^2) \\& & &= 200.96\end{aligned}$$

$$\begin{aligned}V &= A_{\text{base}} \cdot h \\&= (200.96)(20)\end{aligned}$$

$$V = 4019.2 \text{ ft}^3$$

10. **Explain** The illustration shows a cylindrical can and a rectangular box. If the can in the drawing is full, is it possible to pour the contents into the box without the box overflowing?



$$\begin{aligned}\textcircled{1} \text{ can } V &= (\pi r^2)(h) \\&= (3.14)(5^2)(12) \\V &= 942 \text{ in}^3\end{aligned}$$

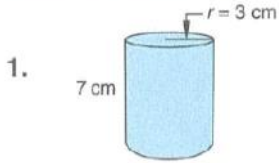
$$\begin{aligned}\textcircled{2} \text{ Box } V &= LHW \\&= (12)(15)(5) \\V &= 900 \text{ in}^3\end{aligned}$$

∴ No. The volume of the box is smaller than the volume of the can.

Examples #3

Saturday, August 10, 2024 7:27 PM

Find the volume of each figure. Use 3.14 for π , and round your answer to the nearest tenth if necessary.



$$\begin{aligned} V &= (\text{area of Base})(h) \\ &= \pi r^2 h \\ &= (3.14)(3^2)(7) \\ &= 197.82 \end{aligned}$$

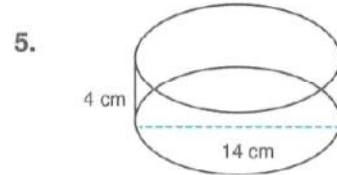
$$V = 197.8 \text{ cm}^3$$



(Hint: The base is a square.)

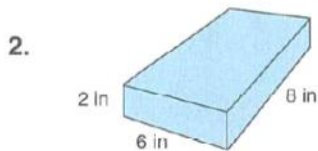
$$V = s^3 = 3^3$$

$$V = 27 \text{ m}^3$$



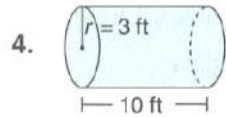
$$\begin{aligned} V &= (\text{area of Base})(h) \\ &= \pi r^2 h \\ &= (3.14)(7^2)(4) \\ &= 615.44 \end{aligned}$$

$$V = 615.4 \text{ cm}^3$$



$$\begin{aligned} V &= LHW \\ &= (2)(6)(8) \end{aligned}$$

$$V = 96 \text{ in}^3$$

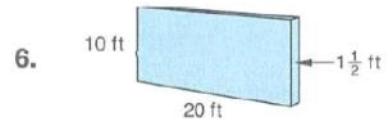


$$V = (\text{area of Base})(h)$$

$$= \pi r^2 h$$

$$= (3.14)(3^2)(10)$$

$$V = 282.6 \text{ ft}^3$$



$$\begin{aligned} V &= LHW \\ &= (10)(20)(\frac{3}{2}) \end{aligned}$$

$$V = 300 \text{ ft}^3$$

VOLUME OF PYRAMIDS, CONES, AND SPHERES

Every solid figure has volume. Remember, when you find volume, you are finding out how many cubes of a certain size would fit into the figure. Some shapes would be impossible to fill with cubes, so you need to use a formula to find the volume.

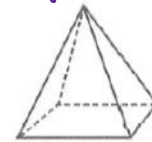
A pyramid has a base with straight sides and triangular faces that meet at a single vertex. The most common pyramid has a square base, but the base can be any polygon.

A cone has a circle for a base and one vertex.

A sphere is a three-dimensional object where each point on the surface is an equal distance from the center. Like the earth.

The GED math formula sheet has the formulas for finding the volumes of these figures. You don't need to memorize them, but take time to understand how they work.

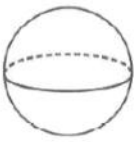
Square
Pyramid



pyramid



cone



sphere

Examples #4

Saturday, August 10, 2024 7:29 PM

Volume (V) Formulas

Pyramid $V = \frac{1}{3}Bh$, where B = area of the Base and h = height

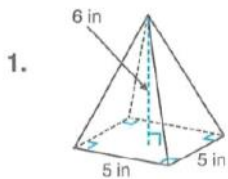
Cone $V = \frac{1}{3}\pi r^2 h$, where r = radius and h = height

Sphere $V = \frac{4}{3}\pi r^3$, where r = radius

on GED, do what the instructions

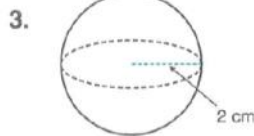
To solve volume problems, substitute the given values and solve. Use 3.14 for π , or the π key on your calculator. On drawings involving height, the height is the perpendicular distance from the base to the vertex.

Find the volume. Round your answers to the nearest tenth.



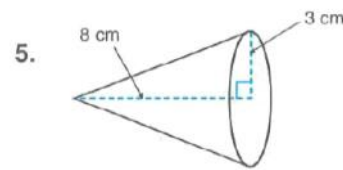
pyramid

$$V = \frac{1}{3}Bh$$



sphere

$$V = \frac{4}{3}\pi r^3$$



cone

$$V = \frac{1}{3}\pi r^2 h$$

Look for connections
How many pyramids
fit in a Rectangular
Prism? $\underline{\underline{3}}$

$$\begin{aligned} V_{\text{pyramid}} &= \frac{1}{3} V_{\text{rect. prism}} \\ &= \frac{1}{3} (bhw) \\ &= \frac{1}{3} (6)(5)(5) \end{aligned}$$

$$V = 50 \text{ in}^3$$

How many cones fit
in a cylinder? $\underline{\underline{3}}$

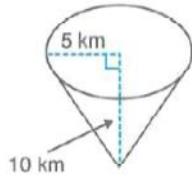
$$\begin{aligned} V_{\text{cone}} &= \frac{1}{3} V_{\text{cylinder}} \\ &= \frac{1}{3} (\pi r^2) h \\ &= \frac{1}{3} (3.14)(3^2)(8) \\ &= 75.36 \end{aligned}$$

$$V = 75.4 \text{ cm}^3$$

$$\begin{aligned} V &= \frac{4}{3}\pi r^3 \\ &= \frac{4}{3}(3.14)(2^3) \\ &= 33.493 \end{aligned}$$

$$V = 33.5 \text{ cm}^3$$

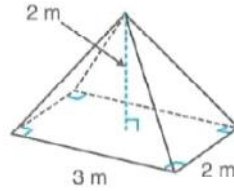
2.



$$\begin{aligned}
 V &= \frac{1}{3} [\pi r^2 (h)] \\
 &= \frac{1}{3} (3.14) (5^2) (10) \\
 &= 261.6
 \end{aligned}$$

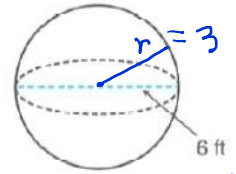
$$V = 261.7 \text{ Km}^3$$

4.



$$\begin{aligned}
 V &= \frac{1}{3} (Lwh) \\
 &= \frac{1}{3} (3) (2) (2) \\
 V &= 4 \text{ m}^3
 \end{aligned}$$

6.



$$\begin{aligned}
 V &= \frac{4}{3} \pi r^3 \\
 &= \frac{4}{3} (3.14) (3^3) \\
 &= 113.04
 \end{aligned}$$

$$V = 113.04 \text{ ft}^3$$

Problem Solving with Volume

Saturday, August 10, 2024 7:32 PM

You can combine solids to form more complex figures.

FINDING THE VOLUME OF COMPLEX FIGURES

Example A cone shape is cut from a foam cylinder to create a form used to dry painted machine parts. If the height of the cylinder is 6 inches and the height of the cone is 5 inches, how many cubic inches of foam are left in the form? Round to the nearest cubic inch. (Hint: The radius of the cylinder is the radius of the cone plus the 1-inch thickness of the foam.)

Step 1

Find the volume of both figures.

Cylinder: $V = \pi r^2 h$
 $= 3.14 \cdot 3^2 \cdot 6$
 $\approx 169.56 \text{ in}^3$

Cone: $V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \cdot 3.14 \cdot 2^2 \cdot 5$
 $\approx 20.93 \text{ in}^3$

There are **149 in³** left in the foam.



Step 2

Subtract and round to the nearest cubic inch.

$169.56 - 20.93 = 148.63$,
 which rounds to **149 in³**.

CYLINDER

$h = 6$

$r = 3$

$V = (\pi r^2) h$

$= (3.14) (3^2) (6)$

$V = 169.56$

CONE

$h = 5$

$r = 2$

$V = \frac{1}{3} (\pi r^2) h$

$= \frac{1}{3} (3.14) (2^2) (5)$

$V = 20.9\bar{3}$

Volume of Foam left = $169.56 - 20.9\bar{3}$

$V = 148.63$

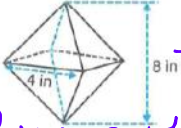
$V \approx 149 \text{ in}^3$

Examples #5

Tuesday, September 3, 2024 12:39 PM

Solve. Use 3.14 for pi.

7. A candy company combines two identical **square pyramids** to make a box for its premium chocolates. The drawing shows the dimensions of the box. What is the **volume to the nearest cubic inch?**

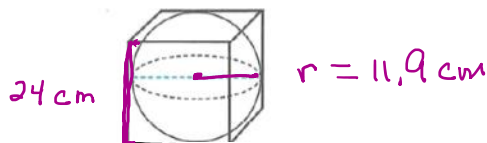
Volume =  4 in

2 ($\frac{1}{3}$ volume of a square prism)

$$\begin{aligned} &= 2 \left(\frac{1}{3} \right) (s^3) \\ &= 2 \left(\frac{1}{3} \right) (4^3) \\ &= 128/3 \\ &= 42.67 \end{aligned}$$

$V = 43 \text{ in}^3$

8. At a sporting goods store, basketballs are sold in boxes. **Each basketball has a radius of 11.9 centimeters. Every edge of the box measures 24 centimeters.** When the basketball is in the box, how much space is left in the box? Round to the nearest cubic centimeter.



$$\begin{aligned} V &= V_{\text{Box}} - V_{\text{sphere}} \\ &= (s^3) - \left[\frac{4}{3} \pi r^3 \right] \\ &= (24^3) - \left[\frac{4}{3} (3.14) (11.9)^3 \right] \\ &= 13824 - 7055.199 \\ &= 6768.801 \end{aligned}$$

$V = 6769 \text{ cm}^3$

9. A model rocket has the dimensions shown in the drawing. If the rocket has a **diameter of 2 inches**, what is the volume in cubic inches?

$r = 1 \text{ inch}$

Round to the nearest cubic inch.



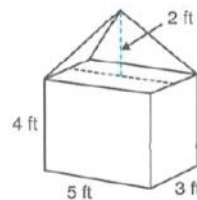
cone is $\frac{1}{3}$ CYLINDER

CYLINDER

$$\begin{aligned} V &= (\text{cone}) (\text{cylinder}) \\ &= \left[\frac{1}{3} \pi r^2 h \right] + \left[\pi r^2 h \right] \\ &= \left[\frac{1}{3} (3.14) (1^2) (6) \right] \\ &\quad + \left[(3.14) (1^2) (10) \right] \\ &= (6.28) + (31.4) \\ &= 37.68 \end{aligned}$$

$V = 38 \text{ in}^3$

10. Max plans to build a storage shed with the dimensions shown in the drawing. **What is the volume of the shed in cubic feet?**



PYRAMID

+

Rectangular Prism

$$\begin{aligned} V &= \frac{1}{3} Bh + Bh \\ &= \frac{1}{3} [(5)(3)] 2 + [(5)(3)] 4 \\ &= 10 + 60 \end{aligned}$$

$V = 70 \text{ ft}^3$